



الهيئة الاتحادية للهوية
والجنسية والجمارك وأمن المنافذ
FEDERAL AUTHORITY FOR IDENTITY,
CITIZENSHIP, CUSTOMS & PORT SECURITY

ID Card Toolkit Android Sample Build Instructions

Version 1.4

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Revision History

Version	Date	Description of Changes
1.0	29th May, 2018	Release Version 1.0.5
1.1	15th Jun, 2018	Release Version 1.0.6
1.2	07th Jul, 2018	Release Version 1.0.7
1.3	20th Dec, 2023	Release Version 3.0.0
1.4	Mar 2026	Toolkit v3.0.5. Updated for Android Studio 2024+. Improved plugin instructions.
1.5	Apr 2026	Toolkit v3.1.0.
1.6	May 2026	Toolkit v3.1.6. Updated toolchain matrix; added AndroidManifest configuration; added pointer to Service Provider Update Instructions for migration deltas.

Abbreviations

Abbreviation	Description
AAR	Android Archive
ABI	Application Binary Interface
ADB	Android Debug Bridge
ICP	Federal Authority for Identity, Citizenship, Customs & Port Security
IDE	Integrated Development Environment
SDK	Software Development Kit

1. Introduction

The Federal Authority for Identity, Citizenship, Customs & Port Security (ICP) developed the ID Card Toolkit to address the requirements of service providers (SP) to integrate, into their business applications: Identification, Authentication, Digital Signature and Non-repudiation services, around the capabilities of the Emirates ID Card.

1.1. Purpose

This document is a developer's reference guide to create and build the sample application provided with the ICP ID Card Toolkit Android SDK.

2. Android Sample Application Build Instructions

2.1. Prerequisites

- ICP ID Card Toolkit Android SDK (EIDAToolkit.aar)
- Plugin libraries (.aar files) for the readers and biometric devices your application supports
- Android Studio (latest stable – Hedgehog 2023.1.1 or newer recommended)
- A development host running Windows, Linux, or macOS

The bundled sample project is built and verified with the toolchain in the table below. Older toolchains may still compile but are not actively tested; align your project to these versions if you hit obscure build or DEX-merge errors.

Component	Minimum / verified version
Java / JDK	Java 11 (sourceCompatibility / targetCompatibility VERSION_11)
Android Gradle Plugin	8.2.1
Gradle wrapper	8.14.2 (the bundled gradlew downloads this on first run)
compileSdkVersion	35
targetSdkVersion	34 (raise as Google Play raises the floor)
minSdkVersion	28 (Android 9.0)

2.2. Build Instructions

2.2.1. Step 1: Import the Sample Project

1. Open Android Studio.
2. Select **Open an existing Android Studio project**.
3. Browse to the Toolkit sample project directory and click **OK**.

2.2.2. Step 2: Place Plugin Libraries

Copy or replace the plugin .aar files into the corresponding plugin folders of the imported project.

Note: The Android sample includes integration with 23 hardware plugins covering smartcard readers, NFC readers, and fingerprint scanners. Only include plugins that support the target build architecture; an incomplete set of binaries for a given ABI will cause load failures at runtime.

2.2.3. Step 3: Place Toolkit SDK

Copy or replace EIDAToolkit.aar into the EIDAToolkit folder of the imported project. The Toolkit SDK supports armeabi, armeabi-v7a, and arm64-v8a ABIs.

2.2.4. Step 4: Build

Build the project in Android Studio (**Build > Make Project** or **Ctrl+F9**).

2.2.5. Step 5: Deploy and Configure

1. Connect an Android device via USB and ensure it is detected by ADB.
2. Run the sample project on the device from Android Studio (**Run > Run App** or **Shift+F10**).
3. On the device, a folder named `EIDAToolkit` is created in the device storage. Place the following files in this folder:
 - `config_ap` – application configuration file
 - SP-specific configuration files in the subdirectory specified by the `environment` parameter

For example, if `config_ap` contains `environment = dev`, place SP configuration files in the `EIDAToolkit/dev/` folder.

Note: If `config_ap` is not provided, default configuration is used and all SP configuration files must be placed directly in the `EIDAToolkit` folder.

2.2.6. Step 6: Run

After placing the configuration files, use the running sample application to access the Toolkit services.

2.3. Additional Build Requirements

2.3.1. Gradle `packagingOptions`

The app `build.gradle` must include the following inside its `android { }` block:

```
android {
    packagingOptions {
        doNotStrip "**/*.so"
        pickFirst "lib/arm64-v8a/libc++_shared.so"
        pickFirst "lib/armeabi-v7a/libc++_shared.so"
        jniLibs {
            useLegacyPackaging = true
        }
    }
    defaultConfig {
        multiDexEnabled true
    }
}
```

- `doNotStrip` prevents Gradle from re-stripping the plugins' native libraries during APK packaging; re-stripping would invalidate the Toolkit's plugin integrity verification.
- `useLegacyPackaging = true` makes Android extract `.so` files to the device filesystem at install time so the Toolkit can `dlopen` them. Without it the libraries stay compressed inside the APK and load fails.
- `pickFirst` on `libc++_shared.so` resolves duplicate-library conflicts when two plugin AARs ship `libc++_shared.so` built against different NDK versions.
- `multiDexEnabled true` is required because the Toolkit + plugin set exceeds the 64K method limit.

2.3.2. AndroidManifest application attributes

Two attributes must be present on the `<application>` tag in `AndroidManifest.xml`:

```
<application
    ...
    android:extractNativeLibs="true"
    android:requestLegacyExternalStorage="true">
```

- **extractNativeLibs="true"** – Android Gradle Plugin 4.1 and newer set this to `false` by default. The Toolkit loads its smart-card and biometric plugins through `dlopen()` against a filesystem path; compressed-in-APK libraries return "library not found" at load. This setting and the `useLegacyPackaging = true` setting above must agree.
- **requestLegacyExternalStorage="true"** – Android 10 (API 29) introduced scoped storage. If your app reads `config_*` files from external-storage paths, the flag enables direct path access at `targetSdkVersion = 29`. From Android 11 (API 30) onward Google ignores the flag for most apps; the long-term recommendation is to deploy configuration files into the app's private storage (`getFilesDir()`) instead.

If your application embeds biometric ML models alongside the Toolkit's own and routinely hits the default per-app heap limit, also add `android:largeHeap="true"`. If your Validation Gateway endpoint is reached over plain HTTP rather than HTTPS, add `android:usesCleartextTraffic="true"` or supply an equivalent `network_security_config.xml`.

2.3.3. USB Reader Support

USB reader support requires a `device_filter.xml` resource file, and the Activity handling USB connection must declare `android:exported='true'` with a `USB_DEVICE_ATTACHED` intent-filter. Without this, USB readers will not be detected on Android API 31 and above.

2.3.4. Building the Android SDK from macOS

Note: When building the Android SDK from macOS, the CMake preset must be overridden with `-DVCPKG_HOST_TRIPLET=arm64-osx` because the presets target Windows build environments by default.

3. Upgrading from an earlier Toolkit version

If you are migrating an existing Android integration from a previous Toolkit release (v3.0.x or earlier), additional deltas apply beyond the steps in this document – for example AGP 8.x compatibility for vendored older plugin AARs, plugin SHA256 integrity changes, and the Gson dependency relocation. See [ID_Card_Toolkit_Service_Provider_Update_Instructions](#) (Android section + "Migrating from an earlier Toolkit version" chapter) for the full list.